TABLE 2: Overall performance comparison and minimization of VFCO with DL-EDI

Metric	Low-freq	Avg-freq	High-freq	Event-Triggered	DVA-P	VFCO	Minimization
CST(s)	30.068 ± 2.425	25.026 ± 1.613	16.101 ± 1.761	20.112 ± 2.124	19.642 ± 1.825	18.904 ± 1.577	$\geq 3.8\%$
USDT(s)	1640.050 ± 176.364	1905.603 ± 171.376	2091.083 ± 141.775	1931.202 ± 199.510	1835.420 ± 138.610	1776.703 ± 93.181	$\geq 3.2\%$
Comp.Cost(s)	6.253 ± 0.678	7.738 ± 0.546	9.729 ± 0.673	8.517 ± 0.735	7.842 ± 0.584	7.251 ± 0.317	$\geq 7.5\%$
Comm.Cost(KB)	8.842 ± 0.592	9.761 ± 0.764	12.005 ± 1.192	10.917 ± 0.936	9.680 ± 0.725	9.205 ± 0.387	$\geq 4.9\%$

Note: Minimizations are measured relative to the maximum baseline value in each row.

TABLE 3: Sensitivity analysis and performance of optimization methods under alternative weight combinations $(\omega_1, \omega_2, \omega_3)$

Weights $(\omega_1, \omega_2, \omega_3)$	Optimization Method	Avg. Freq.	CST (s)	USDT (s)	Comp. Cost (s)	Comm. Cost (KB)
(0.58, 0.23, 0.18)	NSGA-II	0.251 ± 0.023	18.586 ± 1.487	1789.233 ± 107.354	7.243 ± 0.507	9.303 ± 0.651
	NSGA-III	0.257 ± 0.023	17.534 ± 1.403	1810.337 ± 108.620	7.268 ± 0.509	9.391 ± 0.657
	MOEA/D	0.246 ± 0.022	19.400 ± 1.552	1771.164 ± 106.270	7.130 ± 0.499	9.262 ± 0.648
	Lyapunov	0.251 ± 0.023	18.336 ± 1.467	1785.715 ± 107.143	7.274 ± 0.509	9.199 ± 0.644
	VFCO	0.248 ± 0.022	18.826 ± 1.506	1776.852 ± 106.611	7.244 ± 0.507	9.184 ± 0.643
(0.33, 0.33, 0.33)	NSGA-II	0.182 ± 0.016	25.681 ± 2.054	1342.105 ± 80.526	5.284 ± 0.370	6.812 ± 0.477
	NSGA-III	0.188 ± 0.017	24.184 ± 1.935	1368.918 ± 82.135	5.358 ± 0.375	6.885 ± 0.482
	MOEA/D	0.177 ± 0.016	26.762 ± 2.141	1325.682 ± 79.541	5.267 ± 0.369	6.724 ± 0.471
	Lyapunov	0.182 ± 0.016	24.830 ± 1.986	1346.892 ± 80.814	5.401 ± 0.378	6.802 ± 0.476
	VFCO	0.179 ± 0.016	25.961 ± 2.077	1332.418 ± 79.945	5.331 ± 0.373	6.745 ± 0.472
(0.20, 0.60, 0.20)	NSGA-II	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
	NSGA-III	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
	MOEA/D	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
	Lyapunov	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
	VFCO	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
(0.20, 0.20, 0.60)	NSGA-II	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
	NSGA-III	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
	MOEA/D	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
	Lyapunov	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
	VFCO	0.167 ± 0.015	28.373 ± 2.270	1215.996 ± 72.960	4.860 ± 0.340	6.240 ± 0.437
(0.40, 0.40, 0.20)	NSGA-II	0.210 ± 0.019	22.250 ± 1.780	1520.480 ± 91.229	6.082 ± 0.426	7.808 ± 0.547
	NSGA-III	0.216 ± 0.019	20.820 ± 1.666	1543.460 ± 92.608	6.204 ± 0.434	7.923 ± 0.555
	MOEA/D	0.205 ± 0.018	23.160 ± 1.853	1503.320 ± 90.199	6.020 ± 0.421	7.791 ± 0.545
	Lyapunov	0.210 ± 0.019	21.690 ± 1.735	1520.910 ± 91.255	6.194 ± 0.434	7.795 ± 0.546
	VFCO	0.207 ± 0.019	22.443 ± 1.795	1509.730 ± 90.584	6.076 ± 0.425	7.732 ± 0.541

TABLE 4: Impact of n on the performance metrics with DL-EDI

n	**CST (s)**	**USDT (s)**	**Comp. Cost (s)**	**Comm. Cost (KB)**	**Frequency Determination Time (s)**
4	18.379 ± 1.436	1714.453 ± 150.172	7.445 ± 0.861	9.372 ± 0.994	2.181 ± 0.163
8	18.555 ± 1.830	1718.907 ± 111.353	7.303 ± 0.589	9.565 ± 0.933	2.322 ± 0.118
16	18.904 ± 1.577	1776.703 ± 93.181	7.251 ± 0.317	9.205 ± 0.387	2.710 ± 0.228
32	19.179 ± 1.786	1743.850 ± 99.685	7.090 ± 0.505	9.124 ± 0.653	3.140 ± 0.316
64	19.072 ± 2.155	1794.092 ± 134.893	7.065 ± 0.770	9.003 ± 0.819	3.321 ± 0.193

TABLE 5: Impact of T on the performance metrics with DL-EDI

T **(s)**	**CST (s)**	**USDT (s)**	**Comp. Cost (s)**	**Comm. Cost (KB)**
180	22.640 ± 1.412	1563.129 ± 84.316	4.679 ± 0.358	6.038 ± 0.477
240	20.581 ± 1.664	1660.645 ± 156.739	6.021 ± 0.539	7.673 ± 0.678
300	18.904 ± 1.577	1776.703 ± 93.181	7.251 ± 0.317	9.205 ± 0.387
360	16.956 ± 1.114	1883.682 ± 154.675	8.635 ± 0.886	10.856 ± 1.016
420	15.294 ± 1.804	2005.894 ± 179.072	9.908 ± 1.045	12.454 ± 0.908